

DRAWING LIST FOR TEST PACKAGE

NaOCL-593

PROJECT NO.: 22328D

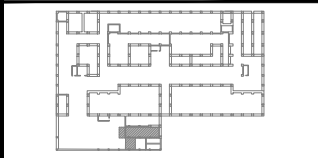
PROJECT TITLE : M15X Ph2 Project

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	593	1FA	OD25-NaOCl-PV3-N-7018-01	1 OF 1	0	CJ-M15W14-PR-A20-000313		
2	593	R2FA	OD25-NaOCl-PV3-E-7018-01	1 OF 1	1	CJ-M15W14-PR-A20-000313		
3	593	2FA	OD25-NaOCl-PV3-E-7018-01	1 OF 1	0	CJ-M15W14-PR-A20-000313		
4	593	3FA	OD25-NaOCl-PV3-7018-01	1 OF 1	0	CJ-M15W14-PR-A20-000313		
5	593	4FA	OD25-NaOCl-PV3-N-7018-02	2 OF 2	0	CJ-M15W14-PR-A20-000313		
6	593	4FA	OD25-NaOCl-PV3-N-7018-01	1 OF 2	1	CJ-M15W14-PR-A20-000315		
7	593	4FA	OD25-NaOCl-PV3-N-7016-01	1 OF 1	0	CJ-M15W14-PR-A20-000315		
8	593	4FA	OD25-NaOCl-PV3-N-7017-01	1 OF 1	0	CJ-M15W14-PR-A20-000315		
9	593	4FA	OD32-NaOCl-PV3-N-7027-01	1 OF 2	0	CJ-M15W14-PR-A20-000315		

14	13	12	11	10	9	8	7	6	5	4	3	2	1
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대외비



KEY PLAN (1F)

NOTE

1. TANK & HWL 미마모 잔존되는 모든 배관은 BLOCK VALVE 설치할 것.
2. 환수 인입지점은 펌프 흡입측에서 최대한 아래쪽에 설치.
2.2m 이상 높이는 곳에 설치하면 별도의 CHAIN VALVE 또는 PLATFORM 구성할 것.
3. MANHOLE : 2개 이상 설치한다.
MANHOLE COVER : 네개
5. 온전치 CLOSE 되고는 배관에는 LOTO 설치.
6. 온전치 PUMP
MAT'L 크로 : CASING / IMPELLER
7. TANK 내 유압, 육안 확인이 가능하고, 비산되지 않도록
회소 길이로 변경할 것. (SLAB과 비동 100 mm 밀착 및 것)
8. 내부 FRP LINING TANK의 SLEEVE는
SUS304 + FRP LINING CHECK
9. I-7007
(W14-P-4711 A/B/C AUTO STOP)
: REMOTE STATUS에서 W14-T-4708 LEVEL
(I-W7008) LOW LOW STOP 또는
RO can the SEQUENCE에 따라 온전치.
(DETAIL 사항은 주부 VENDOR 선정 이후 확립됨).
10. I-7008
(W14-P-4713 A/B AUTO STOP)
: REMOTE STATUS에서 W14-T-4708 LEVEL
(I-W7008) LOW LOW STOP.
11. I-7016
(W14-P-4712 A/B AUTO STOP)
: REMOTE STATUS에서 W14-T-4708 LEVEL
(I-W7008) LOW LOW STOP 또는
RO can the SEQUENCE에 따라 온전치.
(DETAIL 사항은 주부 VENDOR 선정 이후 확립됨).
12. 스기 가동시에 PUMP SUCTION HEADER의
AIR 제거용 VALVE.
13. 공급 : VENDOR / 설치 : SKEP
14. UF/RO/ROCLP에 의해서 온전치면,
발동되는 PLC는 공급되지 않음.

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.10.31
수정 REV.	설명 DESCRIPTION	날짜 DATE

제 도 이 인 학

검 토 한 진 회

승 인 박 동 영

제 작 사

SK 에코플랜트

공 사 명
PROJECT TITLE

M15X Ph4 WWT Project

드면면

DWG. TITLE
PIPING & INSTRUMENT DIAGRAM

REUSE PUMP

출처	NONE
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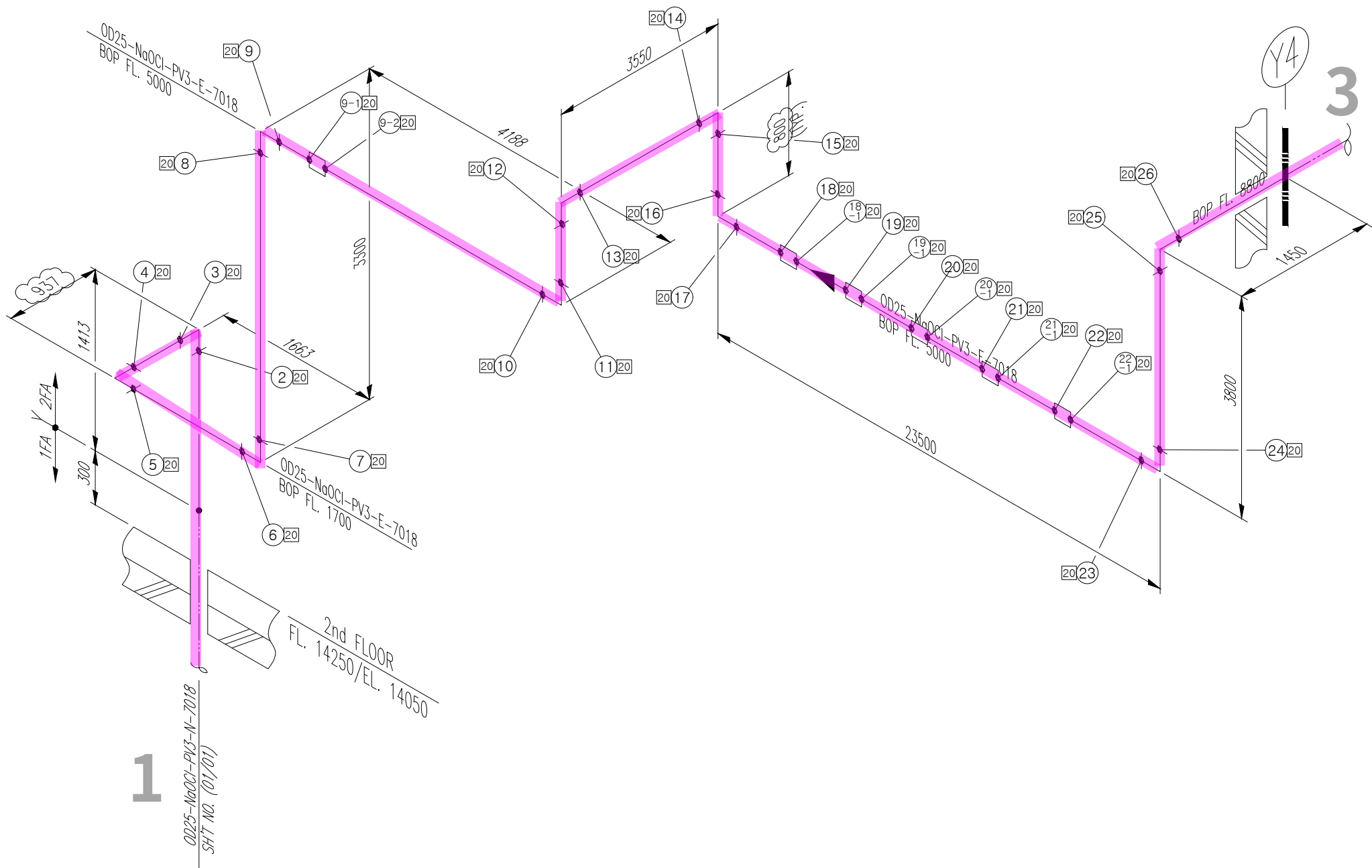
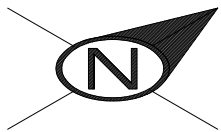
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나. 파	

DATE 2025. 08. 14

PROJECT NO.	22530D	REV.
도면번호	01. MATHWA - BB - 100-000010	1A

SHEET NO.	03-MISN14-FR-A20-000515	1A
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P&ID FOR REF



<NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST



ITEM	SIZE	Q'TY	DESCRIPTION
PV3			
P	OD25	44.6M	PIPE, CPVC, PE, SOCKET FUSION, SDR13.5
E90	OD25	10EA	ELBOW 90 DEG, CPVC, PE, SOCKET FUSION, SDR13.5
INC	25tXOD25	40.0M	관 보존재(고무발포) + 알루미늄 CASING
SC	OD25	6EA	SOCKET, CPVC, PE, SOCKET FUSION, SDR13.5

CLIENT



2

CONTRACTOR	PROJECT NAME
SK ecoplant	M15X Ph4 Project
COMMODITY	JOB NO.
NaOCl	22330D
LINE NO.	SH'T NO. REV.
OD25-NaOCl-PV3-E-7018	01 / 01 1

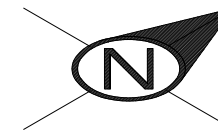
* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	25	DESIGN	10	60	UNIT	R2FA								M15X Ph4 Project	
			OPERATING	7.5	AMB	REF. PLAN DWG.			25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-020016-C			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영		22330D	
						REF. P & ID		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000313								OD25-NaOCl-PV3-E-7018	01 / 01	1



CLIENT  3

CONTRACTOR 		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY NaOCl		JOB NO.	
		22330D	
LINE NO.	0D25-NaOCl-PV3-N-7018	SH'T NO. 01 / 01	REV. 0



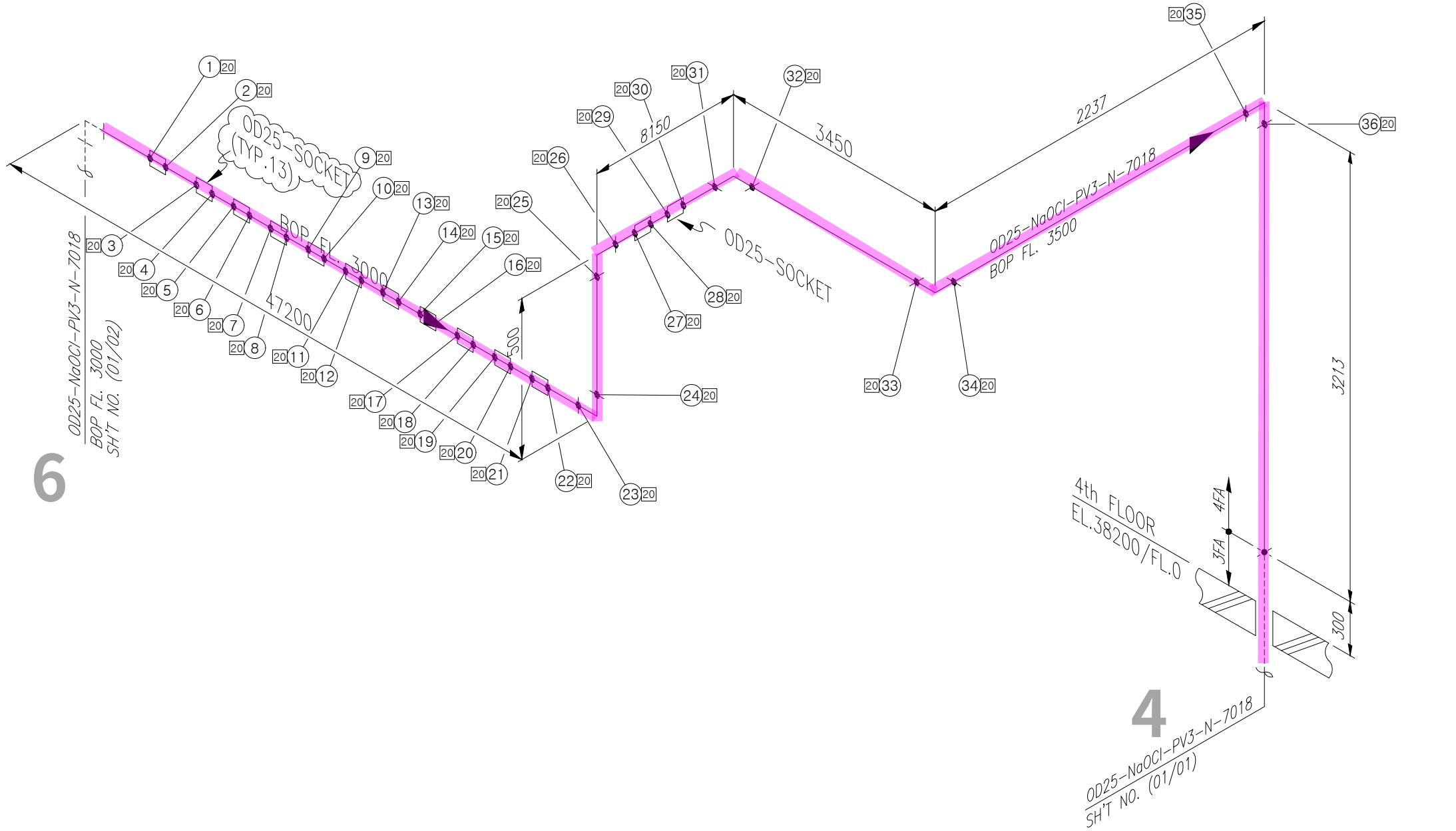
3



4

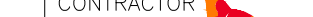
SK ecoplant

6



<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	3.3	60	UNIT	4FA								M15X Ph4 Project	
			OPERATING	3.0	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-040010-B							COMMODITY	NaOCl	JOB NO.
						REF. P & ID	CJ-M15W14-PR-A20-000315		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	OD25-NaOCl-PV3-N-7018	SH'T NO.
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D		02 / 02	0

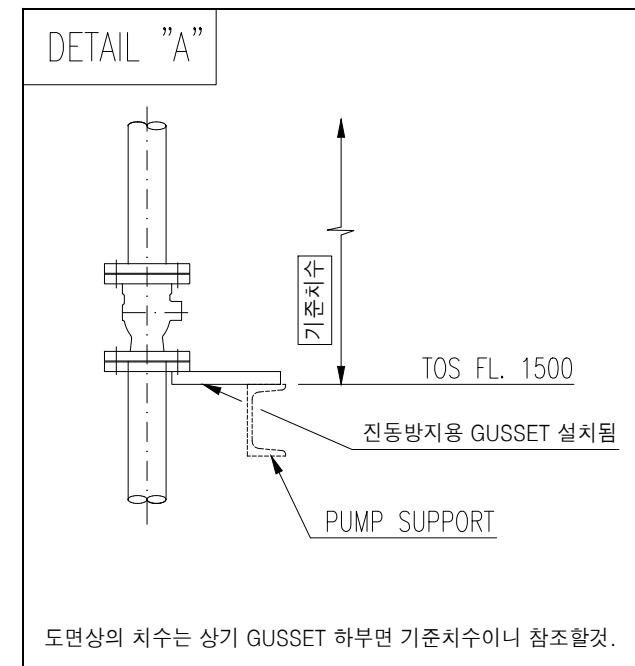
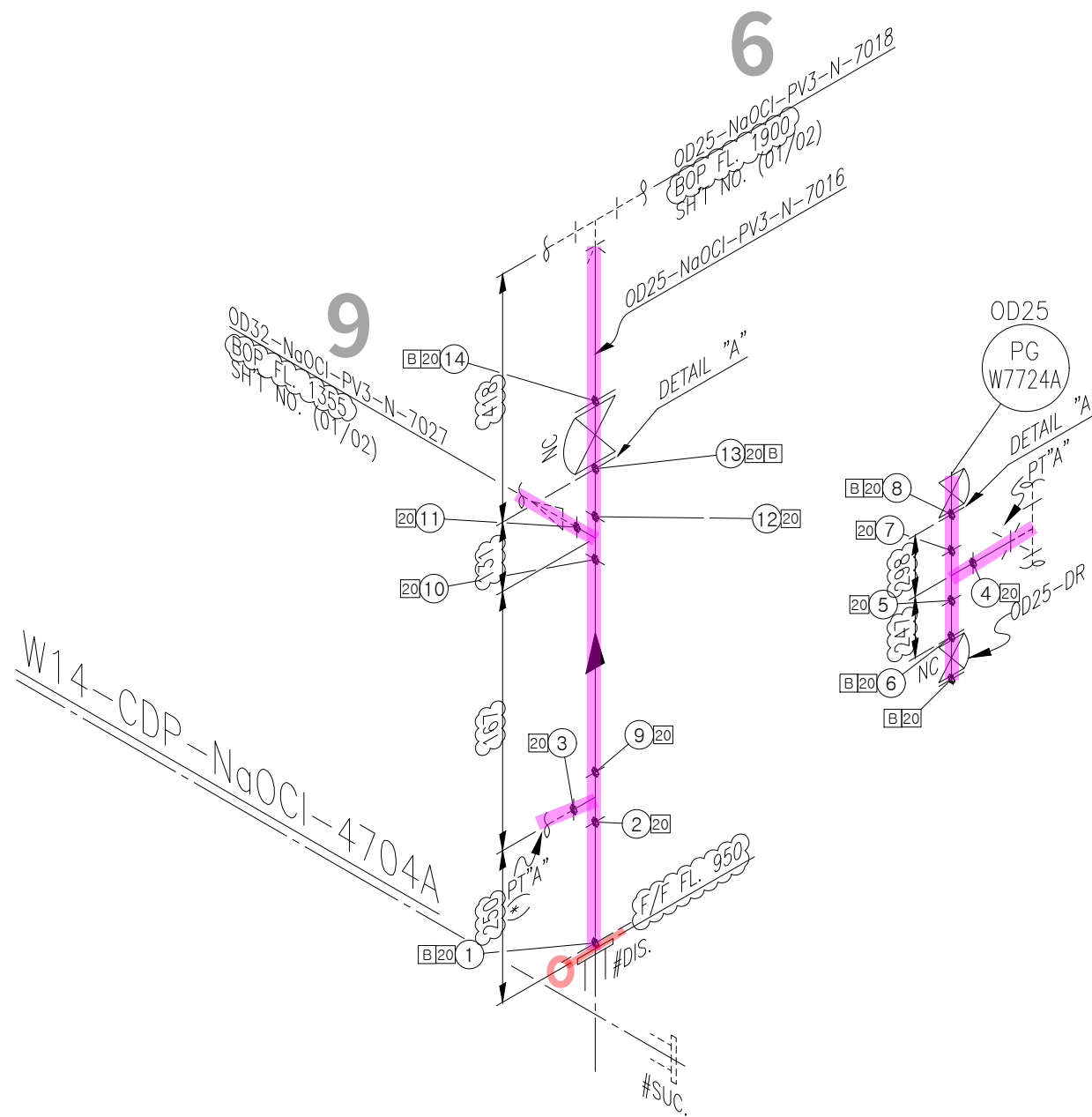


CLIENT  6

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	3.3	60	UNIT	4FA								M15X Ph4 Project		
			OPERATING	3.0	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040010-A			25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	JOB NO.		
									25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	NaOCl	22330D		
						REF. P & ID CJ-M15W14-PR-A20-000315		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	0D25-NaOCl-PV3-N-7018	SH'T NO. 01 / 02	REV. 1



ITEM	SIZE	Q'TY	DESCRIPTION
PV3			
P	OD32	1.3M	PIPE, CPVC, PE, SOCKET FUSION, SDR13.5
T	OD25	3EA	TEE EQUAL, CPVC, PE, SOCKET FUSION, SDR13.5
F	OD25	5EA	FLG, KS 10K, FF, STUB FLG CPVC /
			BACKING RING(PP)+INSERT(STEEL), SDR13.5
FB	OD25	1EA	FLG BLIND, KS 10K, FF, CPVC, SDR13.5
G	OD25	6EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
B	OD25	6SET	MACHINE BOLT & NYLON NUT :STS 304 / W WASHERS
VD	OD25	3EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-CPVC /
			TRIM-PDDM, BB, HW
SH	OD25	6EA	SHIELD
LOTO	OD25	1EA	LOTO
PG	OD25	1EA	PRESSURE GAUGE, KS 10K, FF



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1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR 	PROJECT NAME	
	THICK.	-	DESIGN	3.3	60	UNIT 4FA								M15X Ph4 Project	
			OPERATING	3.0	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040010-A	0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY NaOCI	JOB NO.	
					REF. P & ID CJ-M15W14-PR-A20-000315										
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO. OD25-NaOCI-PV3-N-7016	SH'T NO. 01 / 01	REV. 0



CLIENT  8

